

How does the temperature affect the dynamic viscosity of liquid glycerol at rest, measured by the falling ball method?

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Introduction

Glycerol is used as a thickening agent, humectant and sweetener in foods and beverages. When heated, it is observed that the viscosity of glycerol decreases. It is suggested that as glycerol gets heated up, the energy inside the molecules increases, causing them to overcome the cohesive force binding them easier. These intermolecular forces are behind the viscosity of fluids, so in a fluid of high viscosity, the molecules would be less mobile, as they are attracted to each other.

This investigation explores how the temperature of glycerol affects its viscosity at rest.

Background Information

“Viscosity is a measure of a fluid's resistance to flow” (“DEFINITION of VISCOSITY”). Being viscous causes a liquid to have a sticky and thick consistency, increasing the difficulty to deform that fluid. Within a microscopic level, liquid glycerol ($C_3H_8O_3$) has 3 hydrogen bonds, so the intermolecular bonds are extremely strong. This causes the cohesive forces between the molecules to be very high. As the attractive forces tightly bound the molecules, it is difficult for the molecules to move, causing glycerol to be viscous in room temperature. As the bond strength is inversely correlated to the bond length (Biswal), An increase in temperature would cause the bond strength to weaken, the bond length would then increase. This culminates in the particles being wider apart as the bond decreases in strength significantly

To measure the viscosity of glycerol at different temperatures, the falling sphere method would be used to discover the viscosity of a liquid through finding the terminal velocity of a marble falling. Three forces are exerted on a marble when it falls; gravity, buoyant force and drag force. Stokes law dictates that the drag force on a sphere of radius r moving with a velocity s in a fluid of viscosity η is given by Stokes' law (Vedantu):

$$F_D = 6\pi r\eta s \quad (1)$$

When terminal velocity is reached, which is devised using video analysis, the acceleration is equal to zero as stated by Newton's second law, causing the resultant force, formulated by the forces of buoyancy, fluid drag and weight to be equivalent to zero. The assumption of stokes law is as follows; there is no interference of the marble falling, the flow of the glycerol is laminar, the glycerol is of uniform density, the ball must be sufficiently large, smooth, solid and spherical.

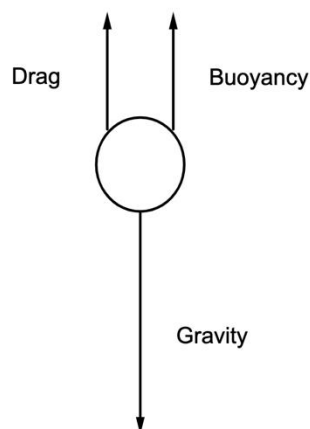


Figure 1: Forces acting on a ball in a viscous liquid in terminal velocity

$$W = F_D + F_B \quad (2)$$

Where W is weight, F_D is fluid drag, and F_B is buoyancy. Substituting the formulas, it becomes

$$mg = 6\pi r\mu s + d_l gV \quad (3)$$

Where m is mass, g is gravitational acceleration, r is the radius of the sphere, η is the viscosity, s is the terminal velocity, d_l is the density of the liquid and V is the volume of the fluid displaced by the sphere (*Buoyancy formula - definition, equations, examples - TOPPR-guides*). Using the formula

$$m = d_p v \quad (4)$$

Where d_p is the density of the sphere, and v is the volume of the sphere (*Learning Chemistry*), m is converted into

$$m = d_p \frac{4}{3}\pi r^3 \quad (5)$$

As V is equal to the volume of the sphere, substituting (5) and volume of the sphere into (3), it becomes

$$d_p \frac{4}{3}\pi r^3 g = 6\pi r\mu s + d_l g \frac{4}{3}\pi r^3 \quad (6)$$

After cancelling π and rearranging (6), the final formula is

$$\mu = \frac{2r^2 g (d_p - d_l)}{9s} \quad (7)$$

To relate temperature to viscosity, the Andrade equation (Frenkel) would be used to fit the graph.

$$\mu = Ae^{\frac{C}{T}} \quad (8)$$

Where A and C are parameters that would be found using the experiment. The literature values of Temperature against Dynamic Viscosity of Glycerol is shown below:

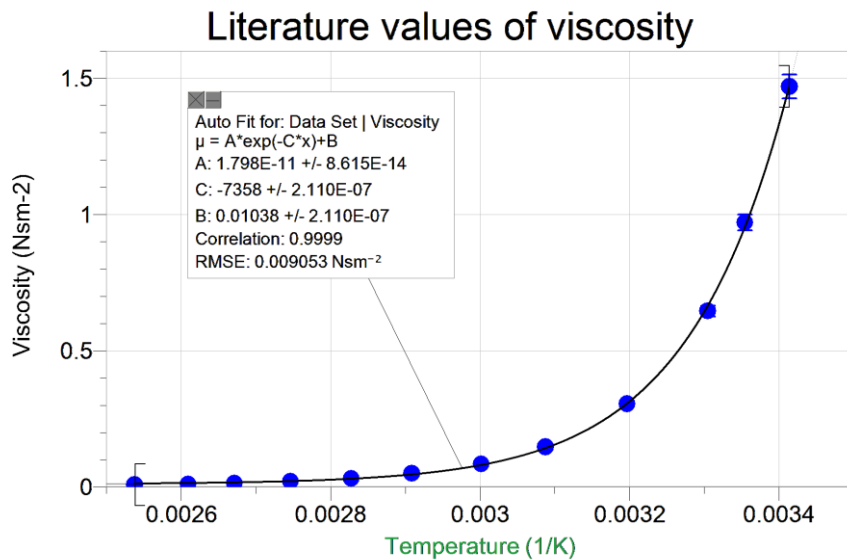


Figure 2: Graph of Literature (Ferreira et al.)

There are many ways to measure viscosity, such as u-tube viscometers, rotational viscometers, and vibrational viscometers. However, the falling sphere method is chosen in the end, due to limitations with apparatus and setup. In addition, the method chosen is the simplest, so there are less controlled variables, making the experiment more accurate.

Pilot Study

Originally, the water would be heated to a certain temperature by an electronic water bath. However, as there were no water baths available in the lab, a beaker with boiled hot water will be used to heat the medium inside a measuring cylinder. Data will be collected when it reaches a certain temperature, until 1 set of data is collected.

In an experiment to determine the range of the independent variable, a temperature range within room temperature and the boiling point of water was considered. As such, temperatures within 25 - 95 °C is considered, with 10 °C intervals. As the room temperature varies with time and weather, recording data in 25 °C is too unreliable. 95 °C is similarly discarded as Newton's law of cooling dictates that the time that the medium stays at 95 °C to be small, and there was also a concern that the medium would not be heated to 95 °C as it has a small window to be heated. Finally, the range of values considered is 35-85 °C.

When a ball initially falls, there would be a small initial window of time that it would accelerate. A marking would be placed at a distance below the surface of the glycerol so that it would have sufficient time that it would reach terminal velocity.

Video analysis would be used so that measurements of the time that the sphere falls in terminal velocity is accurate.

Methodology

	Variable	Detail
Independent Variable	The temperature of glycerol in a measuring cylinder from 35 - 85°C with 10 °C increments. 3 repeats for each temperature.	The temperature is measured by a thermometer with $\pm 0.5^{\circ}\text{C}$ error changed by natural cooling of the water beaker over time
Dependent Variable	The dynamic viscosity of glycerol (μ), in the unit of Nsm^{-2}	The terminal velocity would be determined by video analysis, then calculated by $\mu = \frac{2r^2g(d_p - d_l)}{9s}$
Controlled Variable 1	Density of glycerol	The density of glycerol is determined by the formula $\text{Density} = \frac{\text{Mass}}{\text{Volume}}$ The mass is determined through a balancing scale and the volume determined by measuring cylinder. As the density of glycerol would decrease when the temperature is high, it is found that the density loss is small in literature. More will be elaborated in the evaluation. As the mass of glycerol is found to be 826.42 g and the volume to be 658.26 cm^3, calculated by the formula for the volume of cylinder, the density is calculated to be 1.26 gcm^{-3}

Controlled Variable 2	Density of marble	The density of the marble is determined by the formula $\text{Density} = \frac{\text{Mass}}{\text{Volume}}$ The mass is determined through a balancing scale and the volume determined by finding the diameter and then using the formula of the volume of a sphere. As the mass of the marble is found to be 9.21g and the volume of the 5.58 cm⁻³, the viscosity is found to be 1.65gcm⁻³
Controlled Variable 3	Size of marble	The diameter of the marble is determined through a digital vernier calliper to be 23.00 ± 0.01 mm, and divided by half to get the radius

The experimental setup is as follows:

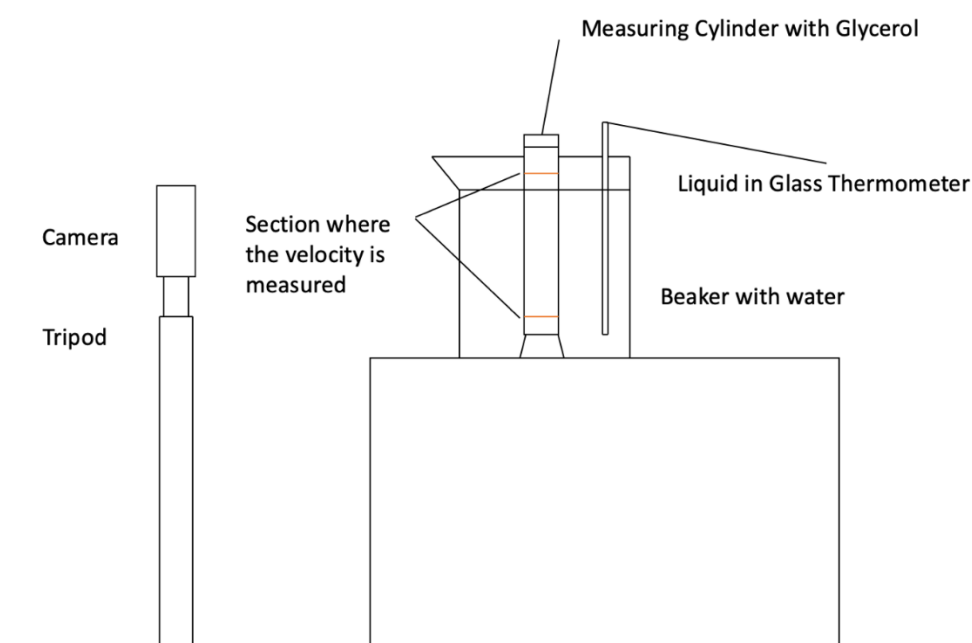


Fig 3: Annotated picture of the experiment setup

List of apparatus:

- 1 Water Boiler
- 2L beaker
- 1 Marble 11.00mm in diameter and 9.21 g
- 1 Camera
- 1 Camera stand
- 1 Liquid in Glass Thermometer with error of $\pm 0.5^{\circ}\text{C}$
- 1 measuring cylinder 29.86mm in diameter
- Glycerol
- Marker
- Metre ruler with error of $\pm 0.05\text{ cm}$
- Digital vernier calliper with error of $\pm 0.01\text{ mm}$
- Measuring scale with error of $\pm 0.001\text{ cm}$

Experimental Procedure

1. Measure and mark the height of glycerol that would be used on the measuring cylinder (23.5 cm) and mark the section that the ball falls in terminal velocity using a metre ruler
2. Measure the diameter of the sphere using a digital vernier calliper and the weight using a measuring scale
3. Boil the water using the water boiler
4. Prepare the camera using a camera stand and place the camera perpendicular to the measuring cylinder to ensure the length of the measuring cylinder on the screen is accurate
5. Slowly pour the glycerol solution into the measuring cylinder until the uppermost marker to prevent air bubbles
6. Put the measuring cylinder into the centre of the beaker to maintain an even temperature throughout the solution
7. Pour the hot water into the beaker to heat up the glycerol
8. Put the thermometer bulb inside the water in about the central location so that the temperature can be recorded
9. When the temperature is close to 85°C , start the camera
10. When the temperature reaches 85°C , release the marble at around the centre of the measuring cylinder and near the top of the glycerol so that it does not hit the walls and reduces the time taken for it to reach terminal velocity
11. After the marble hits the bottom of the measuring cylinder, stop the camera, retrieve the marble, and wipe it clean to remove residue
12. Repeat for temperatures $75 - 35^{\circ}\text{C}$, 3 times for each temperature

Safety, Ethical or Environmental Issues

Gloves and safety goggles are worn in case glycerol or water shoots out their containers, preventing burns. Paper towels and mops are placed close by to wipe water spills to prevent slipping. Glycerol is mixed with water when disposed as it is flammable and viscous. Experiment is conducted in an area without fire hazards to prevent glycerol burning.

Raw Data

Qualitative Raw Data

After the experiment is conducted, it is found that after the marble is dropped, there is a small amount of glycerol that splashes out. At high temperatures, the water in the bottom part of the beaker will be unclear due to convection currents, which may cause the video analysis to be less accurate.

Quantitative data

		Trial 1		Trial 2		Trial 3	
Temperature (T/°C)	Error T/°C	Terminal Speed s 1 / ms ⁻¹	Error $\Delta 1$ / ms ⁻¹ 1	Terminal Speed s 2 / ms ⁻¹	Error $\Delta 2$ / ms ⁻¹ 1	Terminal Speed s 3 / ms ⁻¹	Error $\Delta 3$ / ms ⁻¹ 1
85.0	0.5	3.88	0.0009	3.77	0.0009	3.80	0.0009
75.0	0.5	2.58	0.0018	2.44	0.0012	2.36	0.0015
65.0	0.5	1.48	0.0033	1.42	0.003	1.30	0.0027
55.0	0.5	0.824	0.0048	0.833	0.0051	0.812	0.0046
45.0	0.5	0.451	0.0096	0.474	0.00102	0.440	0.0087
35.0	0.5	0.230	0.00153	0.236	0.0144	0.241	0.00147

Figure 4: Table of Recorded Terminal Speed

Data Processing

Sample Calculation of Viscosity

A sample calculation of the viscosity using equation (8) is shown below utilising the first trial of 85°C.

Substituting the values into equation (7), the formula becomes

$$\mu = \frac{2 \times 0.11^2 \times 9.81(1650 - 1260)}{9 \times 3.88}$$

$$\mu = 0.0265 \text{ Nsm}^{-2}$$

Sample Calculation of error

The formula of the error of viscosity is shown below:

$$\left(2 \times \frac{\Delta r}{r} + \frac{\left(\frac{\Delta m_p}{m_p} + 3 \times \frac{\Delta r}{r} \right) * d_p + \left(\frac{\Delta m_l}{m_l} + \frac{\Delta h}{h} + 2 \times \frac{\Delta c}{c} \right) * d_l}{d_p - d_l} + \frac{\Delta s}{s} \right) \times \eta$$

Where c and h are the radius and height of the viscosity of glycerol in the measuring cylinder. Substituting the values, the formula becomes

$$\left(2 \times \frac{0.01}{22} + \frac{\left(\frac{0.01}{9.21} + 3 \times \frac{0.01}{22} \right) \times 1.65 + \left(\frac{0.01}{826.42} + \frac{0.05}{23.5} + 2 \times \frac{0.01}{29.86} \right) \times 1.26}{1.65 - 1.26} + \frac{0.0009}{3.88} \right) \times 0.0265$$

Where m_p , v_p is the mass and volume of the marble and m_l , v_l is the mass and volume of the glycerol.

The viscosity is calculated below. The average of the terminal speeds of the three trials are calculated.

		Trial 1		Trial 2		Trial 3		Average	
Temperature (T/°C)	Error T/°C	Viscosity 1/ Nsm ⁻²	Error Δ1 / Nsm ⁻²	Viscosity 2/ Nsm ⁻²	Error Δ2 / Nsm ⁻²	Viscosity 3/ Nsm ⁻²	Error Δ3 / Nsm ⁻²	Viscosity Nsm ⁻²	Error ΔNsm ⁻²
85.0	0.5	0.0265	0.000546	0.0273	0.000562	0.0271	0.000557	0.0270	0.000555
75.0	0.5	0.0399	0.000839	0.0422	0.000879	0.0436	0.000915	0.0419	0.000878
65.0	0.5	0.0695	0.00157	0.0724	0.00163	0.0791	0.00177	0.0737	0.00166
55.0	0.5	0.125	0.00327	0.123	0.00327	0.127	0.00330	0.125	0.00328
45.0	0.5	0.228	0.00950	0.217	0.00488	0.234	0.00938	0.226	0.00792
35.0	0.5	0.447	0.0121	0.436	0.0355	0.427	0.0113	0.437	0.0196

Figure 5: Calculated values for viscosity

Curve fitting

A graph showing the relationship between viscosity and temperature is shown here using equation (8). The temperature will be converted to Kelvin and raised to the power of -1.

A graph showing the relationship of Temperature and Viscosity

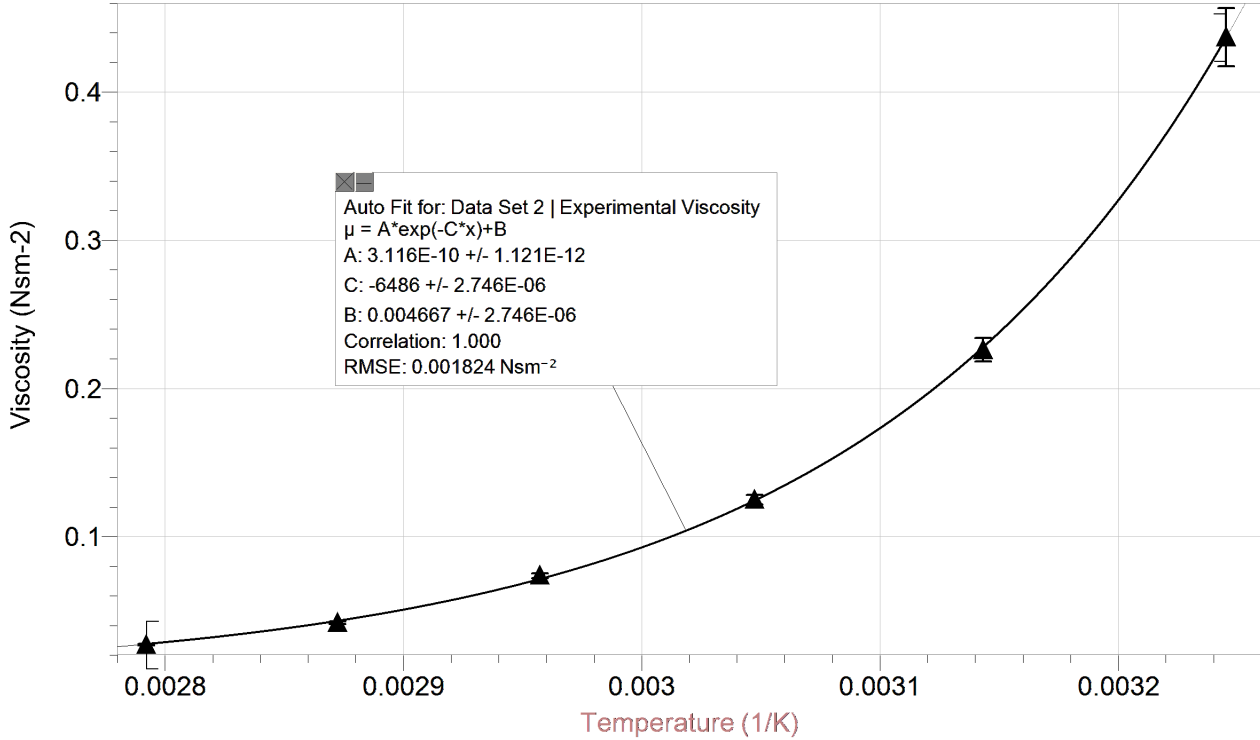


Figure 6: Experimental Values for Viscosity

As the equation relating viscosity and temperature (8) is an exponential function, it will be linearised and shown here:

$$\ln \frac{\mu}{A} = \frac{C}{T}$$

Where A is $3.12\text{E}-10 \pm 1.12\text{E}-12$ and C is $6490 \pm 2.75\text{E}-06$.

To show the relationship between temperature and viscosity, the viscosity and error would be linearised using the natural log function. The formula for the error of $\ln \frac{\mu}{A}$ is shown below:

$$\Delta(\ln(\frac{\mu}{A})) = \frac{1}{2} \left(\ln \left(\frac{\mu}{A} + \left(\frac{\Delta\mu}{\mu} + \frac{\Delta A}{A} \right) \times \frac{\mu}{A} \right) - \ln \left(\frac{\mu}{A} - \left(\frac{\Delta\mu}{\mu} + \frac{\Delta A}{A} \right) \times \frac{\mu}{A} \right) \right)$$

Substituting the values of viscosity for 85 degrees, the error is:

$$\begin{aligned} \Delta(\ln(\frac{\mu}{A})) &= \frac{1}{2} \left(\ln \left(\frac{0.0270}{3.12\text{E}-10} + \left(\frac{\Delta 0.000555}{0.0270} + \frac{1.12\text{E}-12}{3.12\text{E}-10} \right) \times \frac{0.0270}{3.12\text{E}-10} \right) - \ln \left(\frac{0.0270}{3.12\text{E}-10} \right) \right. \\ &\quad \left. - \left(\frac{\Delta 0.000555}{0.0270} + \frac{1.12\text{E}-12}{3.12\text{E}-10} \right) \times \frac{0.0270}{3.12\text{E}-10} \right) \\ \Delta(\ln(\frac{\mu}{A})) &= 3.60\text{E}-03 \end{aligned}$$

For temperature, the formula for error is:

$$\Delta \frac{1}{T} = \frac{\Delta T}{T} \times \frac{1}{T}$$

The values are shown here:

$\frac{1}{T} / \frac{1}{K}$	Error $\frac{1}{T} / \frac{1}{K}$	$\ln \frac{\mu}{A} / \ln \text{Nsm}^{-2}$	Error / $\ln \text{Nsm}^{-2}$
0.00279	3.90E-06	18.3	0.00360
0.00287	4.13E-06	18.7	0.00361
0.00296	4.37E-06	19.3	0.00361
0.00305	4.64E-06	19.8	0.00363
0.00314	4.94E-06	20.4	0.00368
0.00325	5.27E-06	21.1	0.00379

Figure 7: Linearised data values

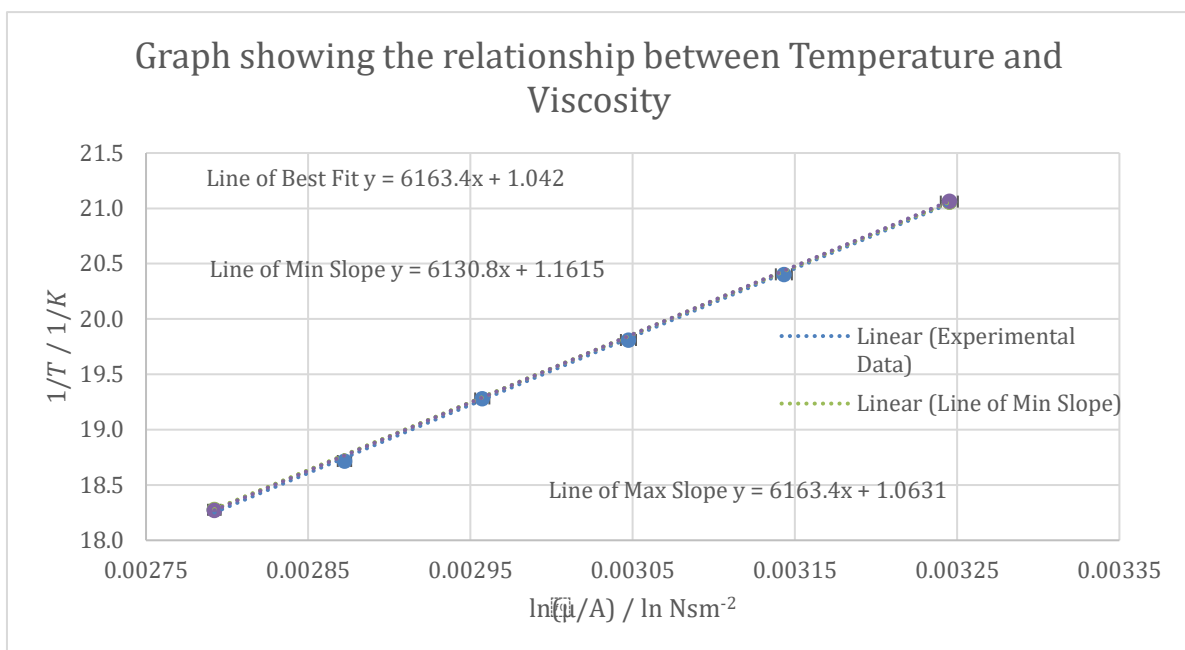


Figure 8: Linearised graph

Conclusion

The best fit line passes through all the error bars. Thus, results of my experiment correlate with the linearised mathematical model, given by the formula $6163.4x + 1.042$, temperature of 35 °C to 85 °C for the conditions stated in my controlled variables. The slope of the experimental result is 6163.4 ± 16.3 and the y-intercept is 1.042 ± 0.0492 . The slope is C, a parameter that was found during the experiment. There is no meaning to the y-intercept as it was created by the linearisation.

Comparison with Literature

Experimental Data

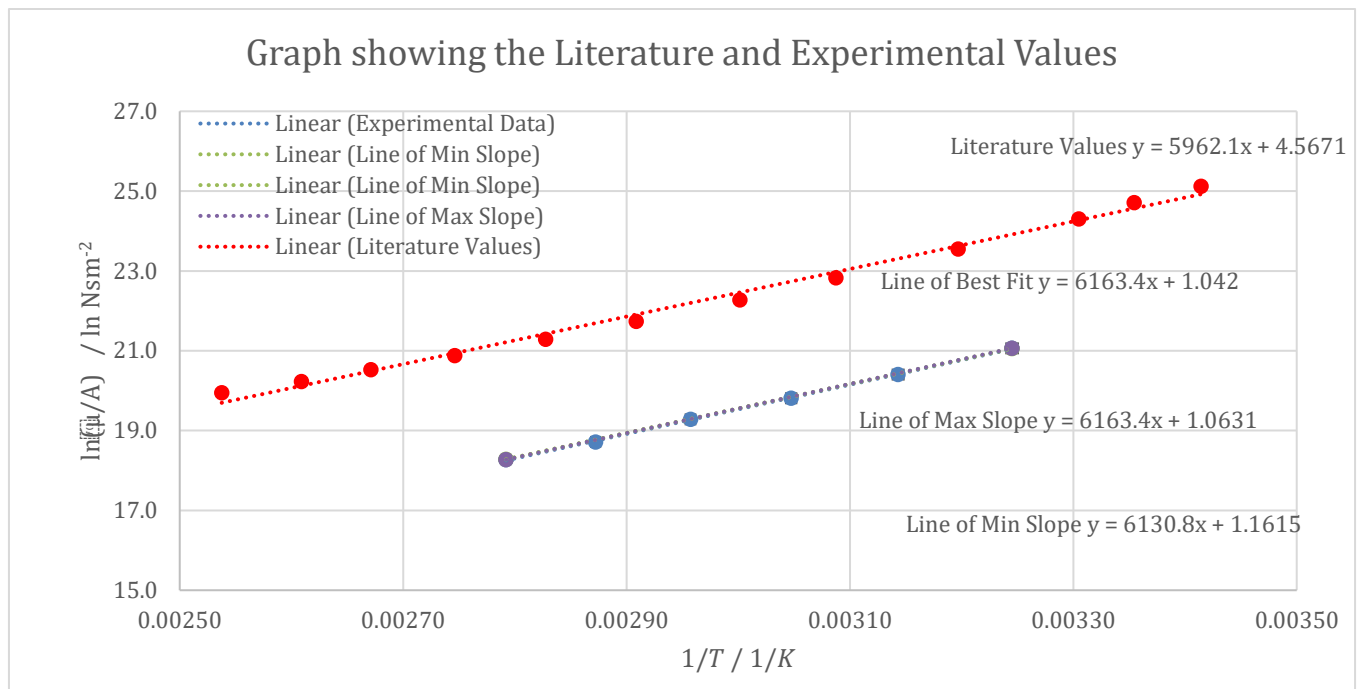


Figure 9: Comparison between literature and experimental values

The slope of the literature values is 5962.1 while the experimental value has 6163.4. The y-intercept of the literature value is 4.5671 while the experimental value has 1.042. Both do not fall under the range of the error. Reasons that the slope of the experimental value is different could be attributed to the continuous change in temperature by the water bath, as the temperature of the glycerol could actually be differing when compared to the recorded value, while the temperature of the literature is constant. In addition, as equation (8) is a formula where A and C are parameters that are found by experimental data, these values are estimated and so the accuracy of the graph could be faulty. Furthermore, as the formula is only usable for liquids, at the boiling point and freezing point of the glycerol, the mathematical model cannot be utilised, making the y-intercept a way to differ the viscosities of various liquids. The variation between the literature and experimental y-intercept could be attributed to the Bernoulli's principle states that as a fluid moves faster, it produces less pressure, and conversely, slower moving fluids produce greater pressure (*Bernoulli's principle - NASA*). In the measuring cylinder, the marble getting closer to the wall would cause the flow speed of the liquid to increase, resulting to a decrease in pressure and thus disrupting the laminar flow, which may change the experimental values of terminal speed, changing the viscosity that was recorded.

Methodology

In the methodological data, it was found that the literature values are found using a Brookfield Thermosel system rotational DV-II+ (model LVDV-II), a rotational viscometer. Rotational viscometers are designed to measure viscosity by analysing the torque required to rotate a spindle submerged in a fluid at a constant

speed(*Rotational viscometer / rotary viscometers*). As such, the methodology of this investigation has a differing method of discovering the dynamic viscosity of the glycerol. In the literature, the uncertainty of each trial is +1%. Utilising 3 trials with different rotational speeds, the combined expanded viscosity uncertainty is 3%. Compared to the experimental uncertainty, the error is much less. As such, it has a higher accuracy.

Sources of Error and Improvement

Error and its effects	Significance and Evidence	Improvements
Change of density as the temperature of glycerol changes	As the glycerol would expand when introduced to heat, the volume would increase. This would cause a systematic error at higher temperatures. A literature value that shows the change in density of glycerol is 0.025gcm^{-3} from 20 - 100°C (Gulyaev and Solonenko), and the change is minimal.	Utilise the changed values of the density of glycerol in calculating the viscosity.
Assumption that the marble must be of uniform density and sufficiently large, smooth, solid, and spherical.	As a singular marble is used to find the value of viscosity, the marble may not satisfy these assumptions	Utilise marble of different densities and average out the values found so that the error can be reduced
Missing the exact moment that the ball falls	As the camera has 30fps when capturing video, the marble would have over 2 frames between markers, reducing the accuracy of the terminal speed	Utilise a camera with higher fps

References

- AZoM. "How Does Temperature Change Viscosity in Liquids and Gases?" AZoM.com, 23 Sept. 2013, www.azom.com/article.aspx?ArticleID=10036. Accessed 17 July 2022.
- "Bernoulli's Principle - NASA." *Bernoulli's Principle*, NASA, 2010, https://www.nasa.gov/sites/default/files/atoms/files/bernoulli_principle_k-4.pdf.
- Biswal, Pragati. "Bond Length and Bond Strength." *Chemistry Steps*, 4 Oct. 2021, <https://www.chemistrysteps.com/bond-lengths-and-bond-strengths/>.
- "Buoyancy Formula - Definition, Equations, Examples - TOPPR-Guides." *Buoyancy Formula*, <https://www.toppr.com/guides/physics-formulas/buoyancy-formula/>.
- "DEFINITION of VISCOSITY." Princeton.edu, 2022, www.princeton.edu/~gasdyn/Research/T-C_Research_Folder/Viscosity_def.html. Accessed 17 July 2022.
- Ferreira, Abel G.M., et al. "The Viscosity of Glycerol." *The Journal of Chemical Thermodynamics*, vol. 113, 2017, pp. 162–182., <https://doi.org/10.1016/j.jct.2017.05.042>.
- Frenkel, J. "The Viscosity of Liquids." *Nature*, vol. 125, no. 3154, 1930, pp. 581–582., <https://doi.org/10.1038/125581b0>.
- Gulyaev, I. P., and O. P. Solonenko. "Hollow Droplets Impacting onto a Solid Surface." *Experiments in Fluids*, vol. 54, no. 1, 2012, <https://doi.org/10.1007/s00348-012-1432-z>.
- Learning Chemistry. "Density - Formula, Definition, Measurement, Calculation." *Learning Chemistry*, 3 Sept. 2022, <https://www.priyamstudycentre.com/2020/06/density-formula.html>.
- "Rotational Viscometer / Rotary Viscometers." *Labcompare*, <https://www.labcompare.com/Laboratory-Analytical-Instruments/870-Rotational-Viscometer-Rotary-Viscometers/#:~:text=Rotational%20viscometers%20are%20designed%20to,allowing%20time%2Ddependent%20fluid%20analysis>.
- Vedantu. "Stokes Law Derivation." *VEDANTU*, Vedantu, 29 Nov. 2022, <https://www.vedantu.com/physics/stokes-law-derivation>.